Using Predicitve Design and Process Insights to Enable Smart Manufacturing in Semiconductor Fabs

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Abstract

Modern semiconductor fabrication pushes the limits of chemistry and physics while simultaneously employing large-scale, cutting-edge processing techniques. While fab expansion and capital expenditures continue to grow, the human element has become ever more demanding and prone to error. To assist with this issue, computer-aided process engineering, process control, and tool monitoring will continue to rise in the coming years. This paper will present the details with results on using Calibre® Fab Insights tool to enable smart manufacturing in three major applications areas within the fab, that can ultimately help in accelerating the yield ramp. First area is process monitoring (PMON), where we leverage usage of design and process features to prescribe product dependent process adjustments which in turn help accelerate NPI [2], improve automated process control (APC), provide actionable insights to identify parameters that most impact any given process step. The second area is tool monitoring (TMON), where the ML model can help early detection of tool drifts, that cause tool to go out of control (OOC), by providing alerts to perform a predictive maintenance. This can reduce the process/tool tuning time to bring them back to spec after a preventive maintenance (PM) cycle. The final area of application highlighted is virtual cross metrology (CM) where a digital metrology twin is created that enables prediction of metrological measurements at every intermediate step along with wafer maps [3,4].

Keywords: Semiconductor Fab Insights, Smart Factory, Machine Learning

1.Introduction

As semiconductor fabrication psuhes the boundaries of physics and chemistry, machine learning (ML) is sought to tackle complex design and process interactions. For advanced nodes, robust process control is essential early in development cycle due to the complexity of the process flow. The foundry faces challenges in high-product-mix manufacturing, requiring increased flexibility in managing diverse customer demands. Coordinating multiple chambers and process steps with different designs and technology nodes is complex as shown in Figure 1. resulting in reduced yields and increased costs.

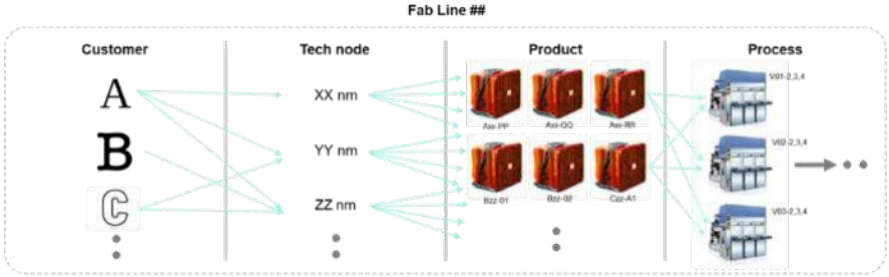

Figure 1. High product mix manufacturing in semiconductor foundry

Additionally, during the first stage of a new product introduction (NPI) in the fab, design validation is a crucial step with only minimal margin of allowed error. Rapid advancements in ML and availability of big data have allowed semiconductor fabs to analyze large amounts of both design and fab data, to make informed decisions and enhance processes through pattern and trend identification.

2.Smart Manufacturing using the Digital Twin approach

Digital twin is popular concept in the industry, where the goal is to allows you to quickly iterate between design and production. The idea that you can very, quickly, virtually prototype and develop new processes, able to characterize every piece of equipment that is in the fab, can really accelerate the transfer between products and production and back from production to product. This concept can further optimize it for the next iterations that fab may face with new designs and product mix by allowing these simulations and validating the different product lines, the complete plans, and serve as a foundation for information exchange between the various groups involved. Digital twins facilitate the exchange of information between all of these different groups, and this explained in this paper, specifically for the semiconductor manufacturing.

2.1Building Digital Twin for Semiconductor Manufacturing

First it is important to identify the fab challenges and the determine how the digital twin can help. It is also critical to identify the infrastructure and resources needed. Bottomline is to increase profit margins, reduce OPEX, CAPEX and increase revenue for any fab. This translates to being able to create product-specific process recipes without test wafers, guarantee highest product quality, with minimal measurements across every wafer, improve yield curve by quick diagnosis of the cause of yield limiting issue, monitor performance of the fleet and predict future performance, for specific products and prevent loss of institutional knowledge with attrition of engineers. The digital twin concept as described earlier can use the huge amount of fab, process and design data to enable addressing the challenges – including monitoring process drift, differentiate between different processes or product mix, monitor tool drift, and restore nominal operation and finally forecast future performance at earlier processing steps in a process sequence.

2.2 Digital Twin Product – Understanding impact of product design

The foundry-fabless model has been very successful, and it has allowed for efficiencies in capital investments and access to the latest manufacturing technology, however as we approach molecular dimensions, understanding the influence of the chip topology becomes relevant. Incorporating design characteristics in semiconductor manufacturing is nothing new, Critical Area Analysis, Resolution Enhancement Techniques like Optical Proximity Correction (OPC) and Source and Mask Optimization (SMO) are a few examples of how design content is used to successfully manufacture semiconductor devices. Examples of the benefits of incorporating design content in recipe setup, process monitoring, and yield analysis, and how incorporating design content is a necessary step to successfully implement machine learning technologies in the manufacturing line has been proved earlier. [4].

2.3 Digital Twin Metrology

Traditional virtual metrology (VM) addresses the trade-off between metrology activities and cost by utilizing data from the process chamber (FDC) to predict metrology results. In addition, design features are extracted and used for prediction across layouts and technologies for high-product-mix manufacturing. Siemens' Calibre® software can be used for feature extraction, and ML models to construct the VM model as a digital twin. Results demonstrate the superiority of the VM model with design features and FDC as

shown previous work [5]. VM can be used at every step of the process to obtain a complete the view of the manufacturing of semiconductor product and estimation of the performance of those devices.

2.4Digital Twin Production – Combining product, process and metrology

Combining design characteristics with process, product and metrology helps arrive to physical models. These models enable to solve many problems like tool and processed drift which results in improved process control and wafer dispositioning. Figure 2. shows the distribution of mean absolute error percentage for representative products for three different technology nodes. We observe a clear reduction in error when the additional design information is included in the model along with the traditional process features.

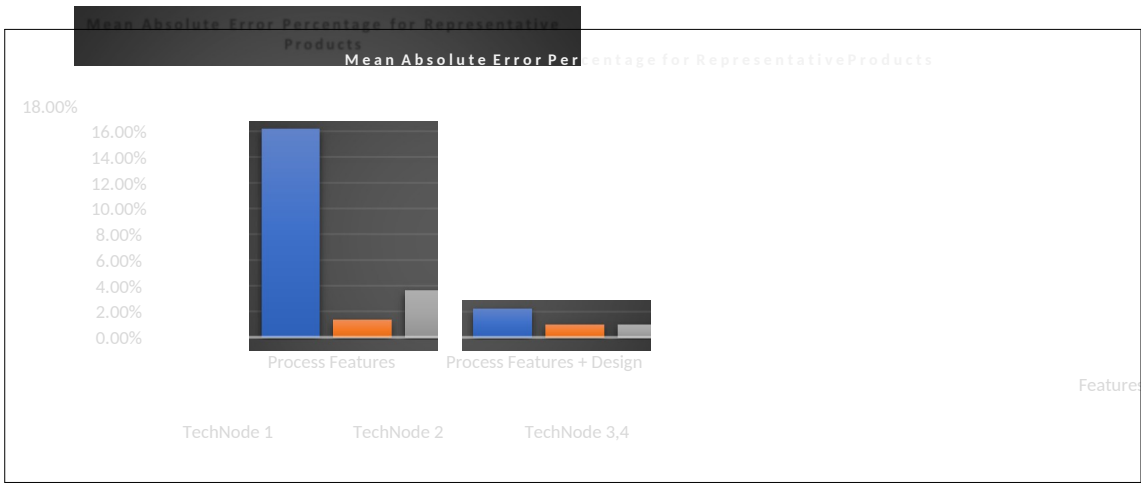

Figure 2. Impact of design features during target prediction for products from multiple technology nodes

3.Results

Calibre® Fab Insights offers a platform to perform data conditioning, ML model building and deployment methodologies to enable a range of fab decision including process and tool monitoring, new product introductions, process recipe optimization, yield/ reliability prediction, identifying yield detractors and preventive maintenance (PM) cycle optimization. Figure 3. shows a flow of the Fab Insight platform.

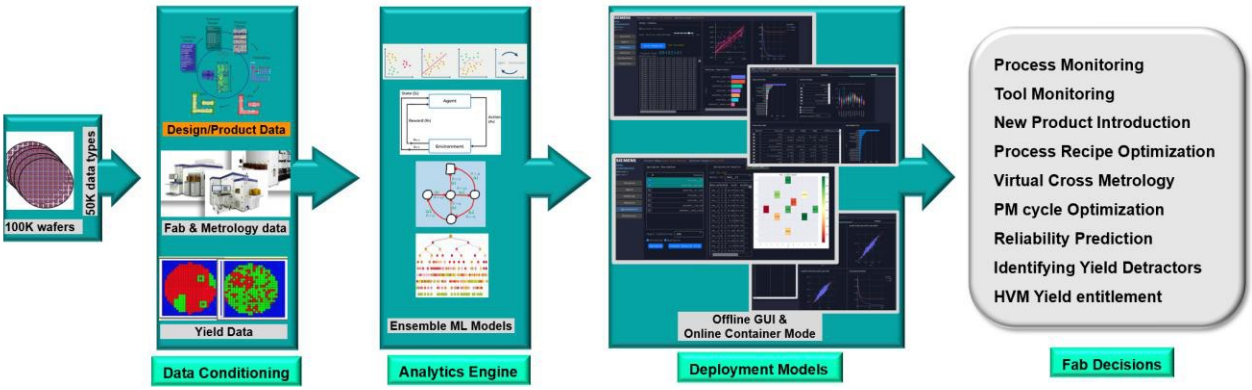

Figure 3. Calibre® Fab Insights Platform

The tool allows both offline GUI based analytics and live container-based deplyoment in the fab. Results of using this technology for three such applications are detailed below.

3.1 PMON – Process MONitoring

Every product reacts differently to the same process conditions so when we have a different mix of products. Under these conditions, when we monitor a specific point, it may feel like either the tools are drifting, or it is due to an interaction between production system and this specific product. Using the PMON application, we are able to differentiate and identify the cause of the process response. One such application of PMON (Figure 4.), is to predict the deposition thickness for any product on a specific Chemical Vapor Deposition (CVD) tool.

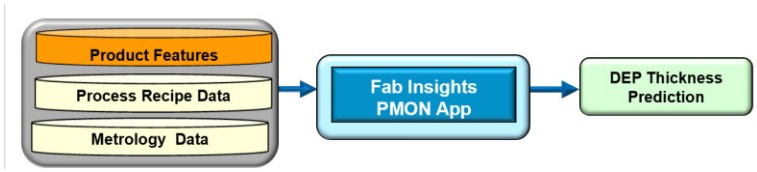

Figure 4. PMON Application used for CVD tool thickness prediction

The model was trained using data (including product, process, and metrology data) from multiple products from four different technology nodes. The resulting Fab Insights PMON model can successfully predict the deposition thickness for an unseen new product with minimal error (Figure 5.) [2].

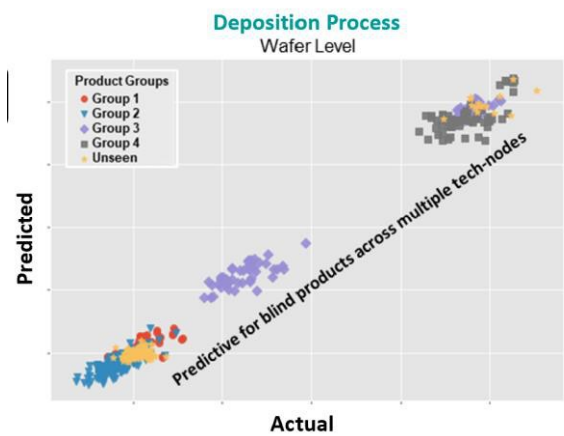

Figure 5. Actual vs. Predicted deposition thickness for various products using PMON model

This will successfully enable to confirm if the problem in the CVD tool is related to a process drift rather than caused by the new product content itself. In which case, the tool doesn't have to be adjusted and it is the recipe that needs to be tuned for the new product.

3.2 TMON – Tool MONitoring

The second application that is explained here is TMON. Consider a CVD and electroplating tool in the traditional fab flow (Figure 6.), where the current process to troubleshoot after OOC or qualification wafers will first require the film thickness parameters to be monitored and measured. Then RCA (Root Cause Analysis) is performed by fab engineers who in turn do manually change the process parameters and then measure the film thickness again for the next set of qualification wafers. This requires several loops to get back the CVD chamber into production.

On the other hand using the TMON for predictive maintenance (Figure 7.) can help use the product, process, tool and metrology data to build an ML model and help not only in alerting the fab on a potential OOC but also provide automatic process control fab recipe adjustments).

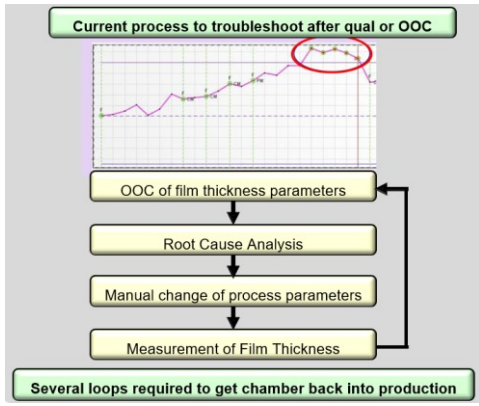

Figure 6. OOC monitoring in a traditional fab flow for a CVD tool

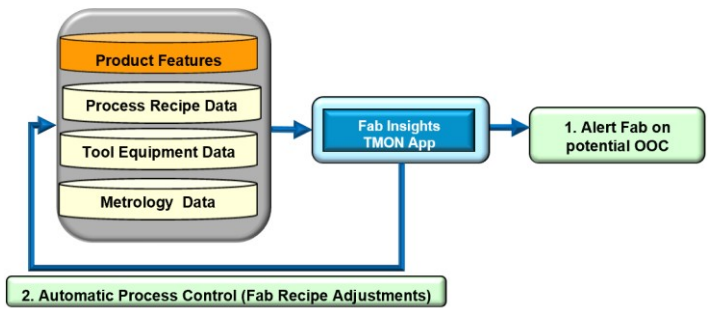

Figure 7. TMON used for predictive maintenance for a CVD tool

We know that tool performance changes with time and need to be monitored across the fleet. If there is a drift caused by a specific tool, that needs to be adjusted. For this purpose, the TMON model trained with CVD fleet data to uncover sub-optimal process parameter settings. Here the ML model can help early detection of tool drifts across the fleet and detect the tool going out of control (OOC) early, by providing alerts to perform a predictive maintenance. This model can also help restore individual nominal tool performance faster by reduce the process/tool tuning time to bring them back to spec after a preventive maintenance (PM) cycle.

3.3 VCM – Virtual Cross Metrology

For the VCM application, we have an example to forecast electrical test performance (e-test) of a back end of line (BEOL) process. This specific process has 42 individual process steps. The VCM model is a sequence of 42 individual process and metrology steps to identify the most significant contributors to e-test. We use the ML model to identify the few critical process steps as “probes” – the steps that have high influence in the final electrical test result.

Figure 8. shows the representation and identification of the 4 probes among the 42 process steps involved for prediction of the electric test target. There is a probe identified as early as process step 6 and another one as late as process step 30, which are indicated to have an impact on the final e-test based on the ML model.

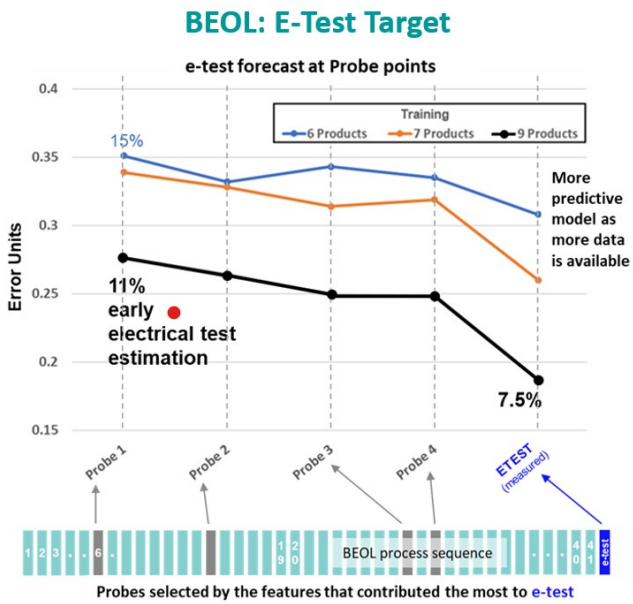

Figure 8. Virtual Cross Metrology (VCM) application results on BEOL E-test target prediction

Multiple models were trained using data from different number of products. In the specific results shared, 3 models were trained with 6,7 and 9 product information included along with the process and metrology data. Using these models at probe 1 (process step 6) we can make an estimation for the final e-test target with a 11-15% error. The error is lower as we try to include a greater number of products in the training data. We observe that with 9 products and at probe 4 (process step 30) – the prediction error for e-test is down to 7.5%. This shows that the model is more predictive as more data is available. This implies that potentially we can continue adding products and try to improve the quality of results.

4.Conclusions

The above results show how the Calibre® Fab Insights platform can be an effective tool to help digital transformation of the semi fabs. Using the concept of digital twins, multiple applications including PMON, TMON and VCM have been implemented and validated to show how it can enable yield ramp impact resulting in higher yield entitlement. Using design data along with existing process and fab metrology is a key differentiating approach highlighted in this paper. Deployment of these technologies in high volume manufacturing (HVM) can be achieved by the concept of using multiple digital threads within the digital twin ecosystem. This also allows the flexibility to allow personalized solutions for every fab and issue. Goal of this tool is to provide the most comprehensive and integrated solutions that are easy to use and deploy and can empower our customers to innovate and drive the digital transformation of the semiconductor fabs.

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